Incentivizing DSR Adoption in UP, India

MIT MANAGEMENT SLOAN SCHOOL

Experimental design for 2025 experiment

Sampling frame

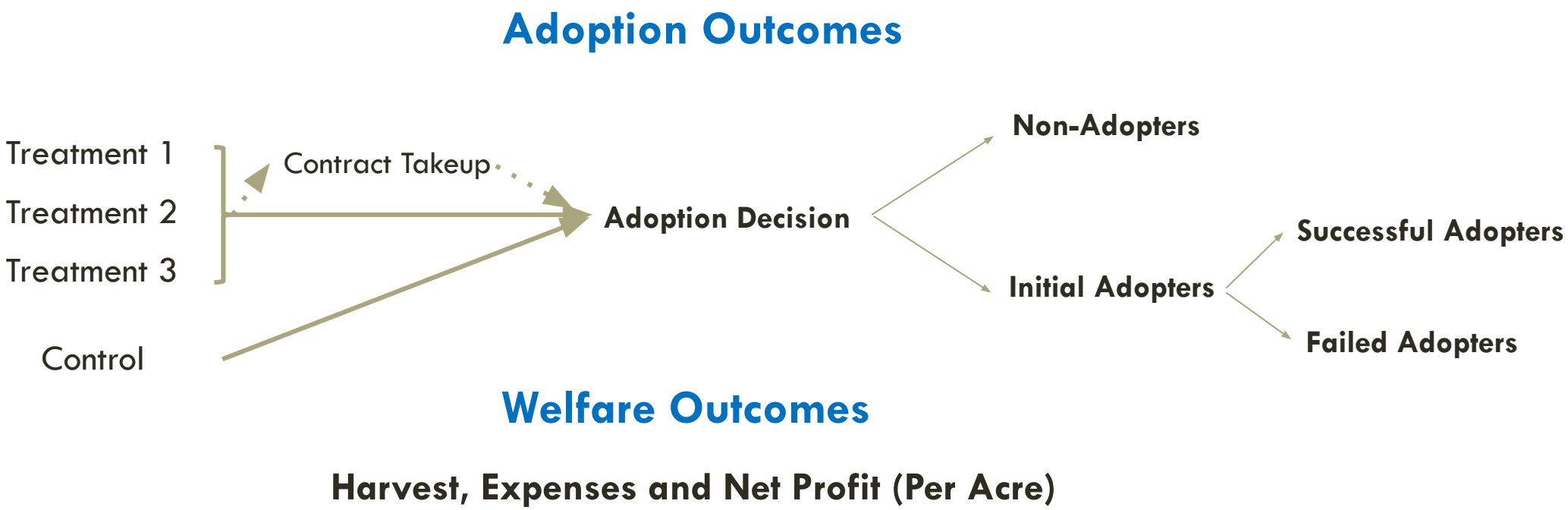
16 villages (UP, India)

Unit of randomization: Village (cluster)

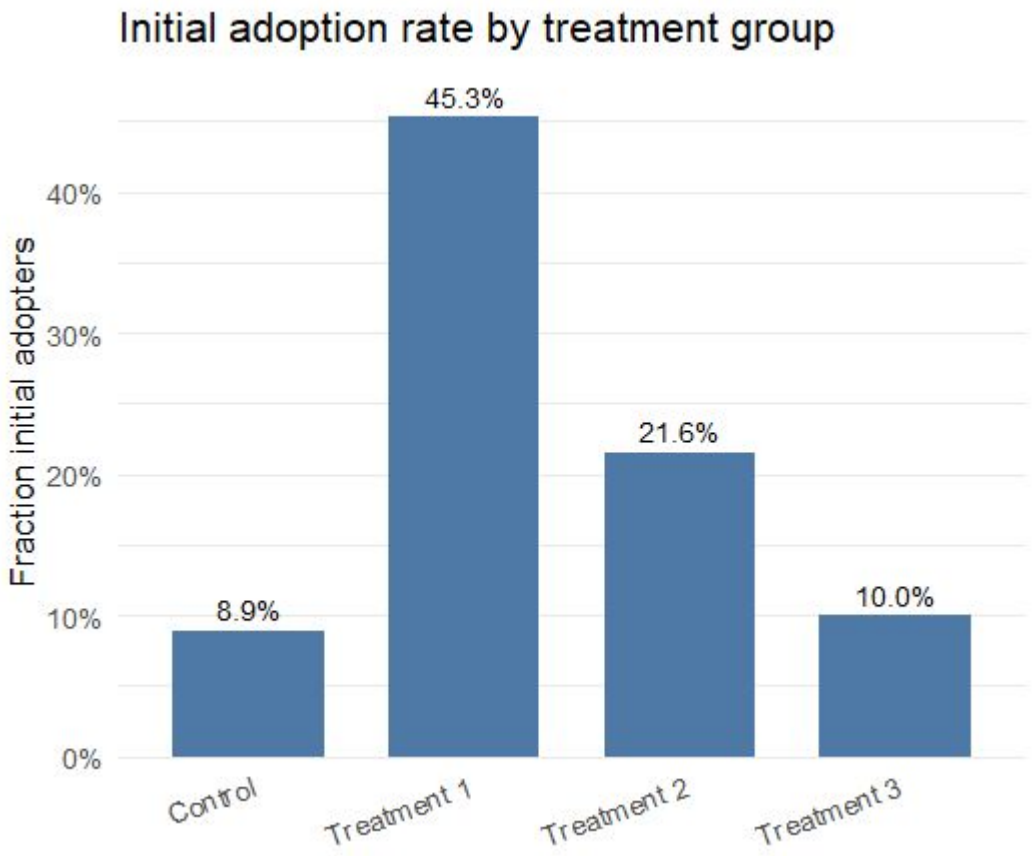
Random assignment

Control	Treatment 1	Treatment 2	Treatment 3
4 villages	4 villages	4 villages	4 villages
60 farmers (4×15)	60 farmers (4×15)	60 farmers (4×15)	60 farmers (4×15)
No financial incentive	**Upfront INR 2000/acre (before sowing; Unconditional)**	**Fixed reward INR 2000/acre (at harvest; Conditional)**	**Price premium INR 100/quintal (at harvest; Conditional)**
15 farmers selected per village	15 farmers selected per village	15 farmers selected per village	15 farmers selected per village

Outcome Evaluation

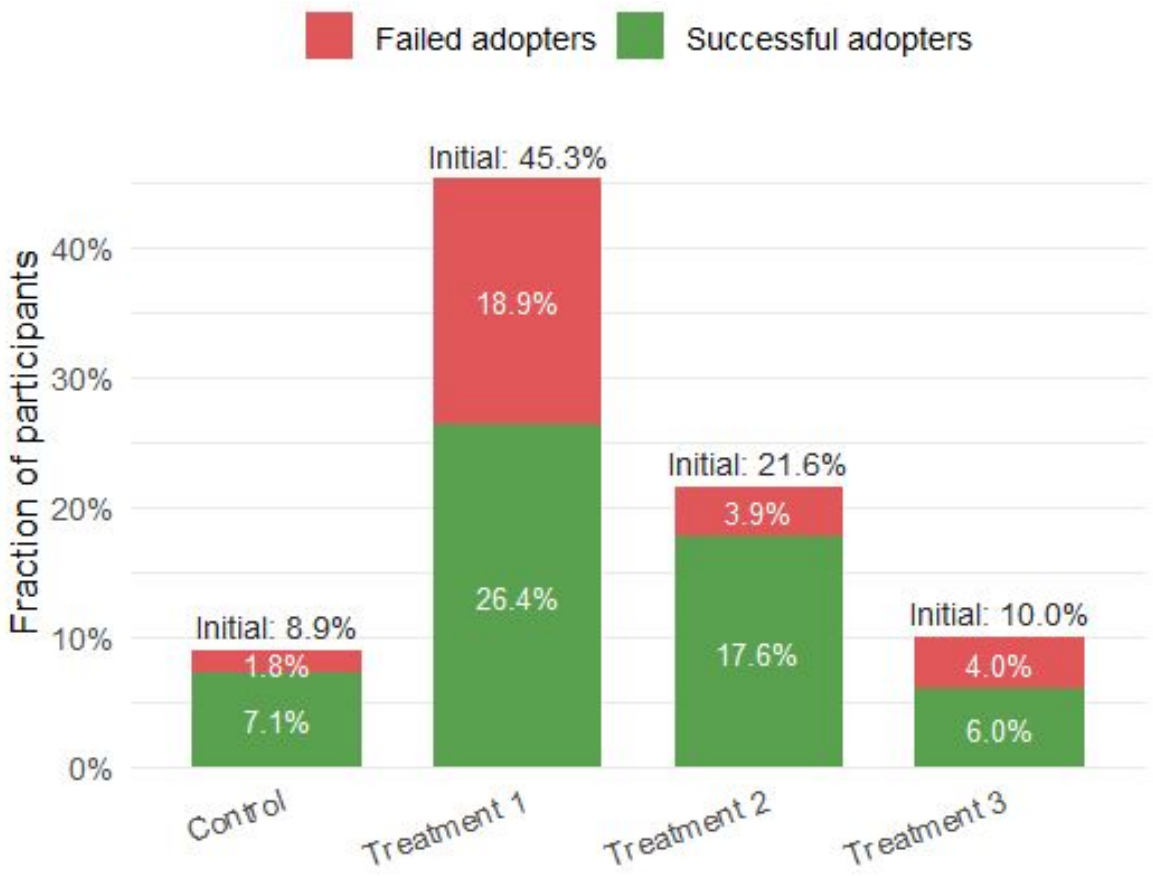


Does offering the incentive motivate farmers to adopt the practice?



- Offering **unconditional upfront payment (T1)** leads to significantly higher initial adoption
- Offering **conditional payment at harvest (T2)** also encourages adoption but is less effective

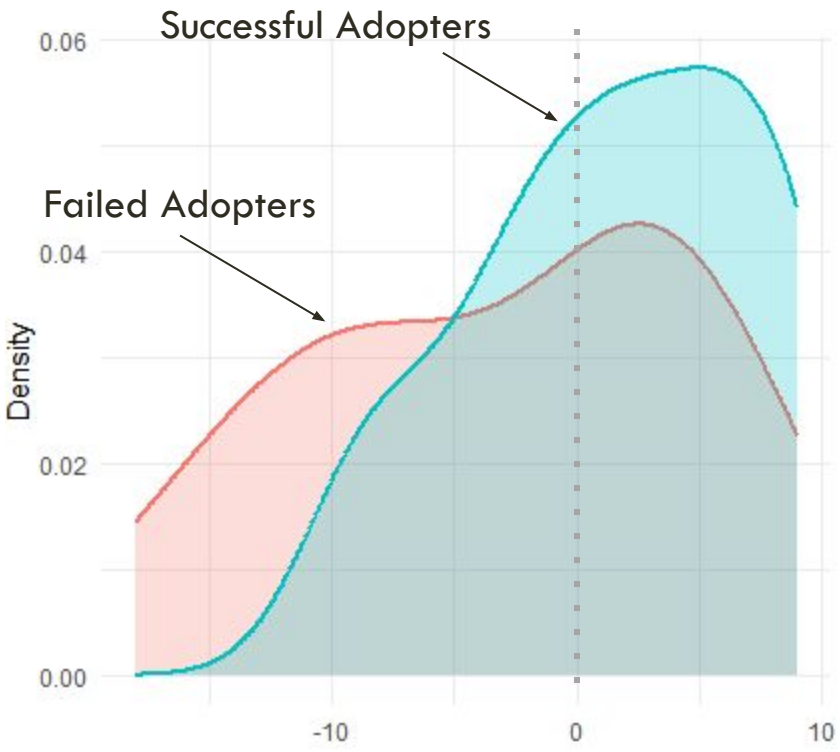
Does offering the incentive lead to **successful adoption?**



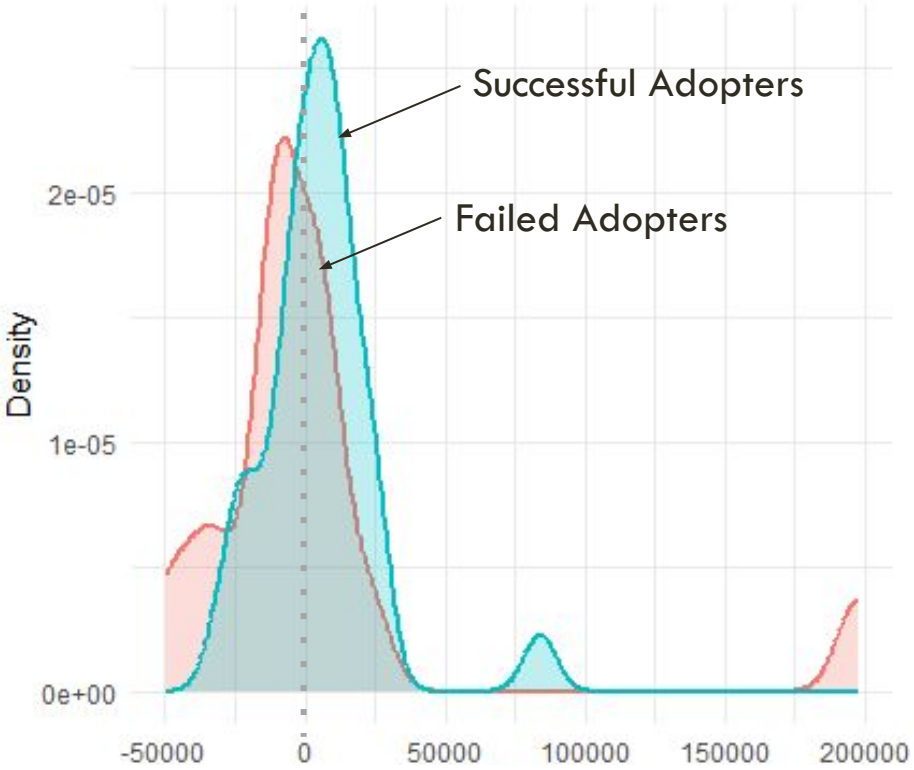
- Offering **unconditional upfront payment (T1)** leads to **higher chance of failed adoption**
- Initial adopters under the **conditional payment at harvest (T2)** have higher success rate

There seems to be an implementation gap: interventions that increase adoption may do so by expanding participation among farmers with **lower implementation capability.**

Failed adoption can be harmful to farmer welfare

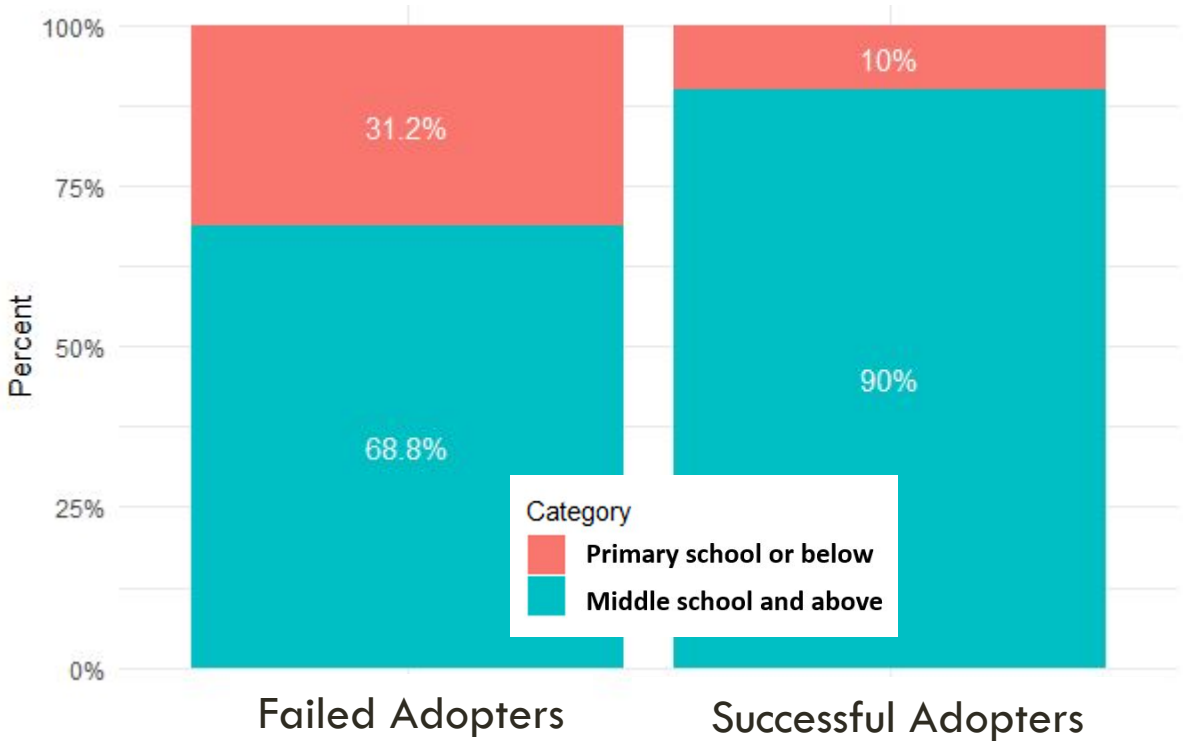


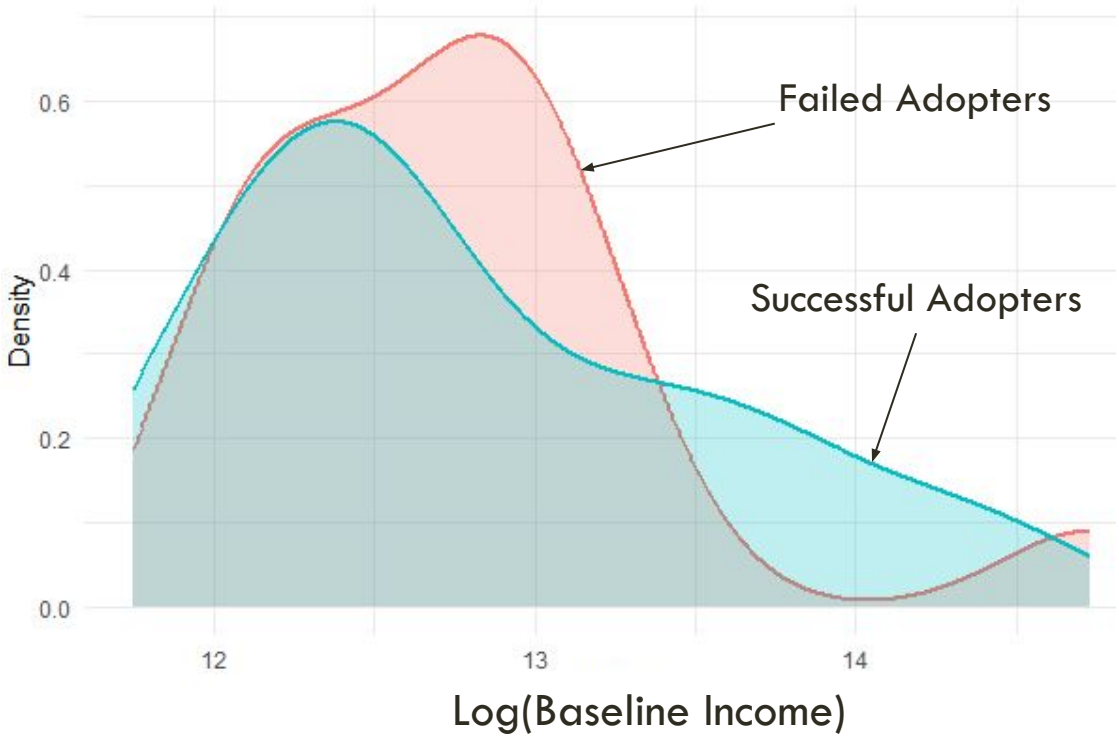
Yield 2025 – Yield 2024



Profit 2025 – Profit 2024

Equity Concern: Failed adopters tend to have **lower education** and **lower baseline household income**





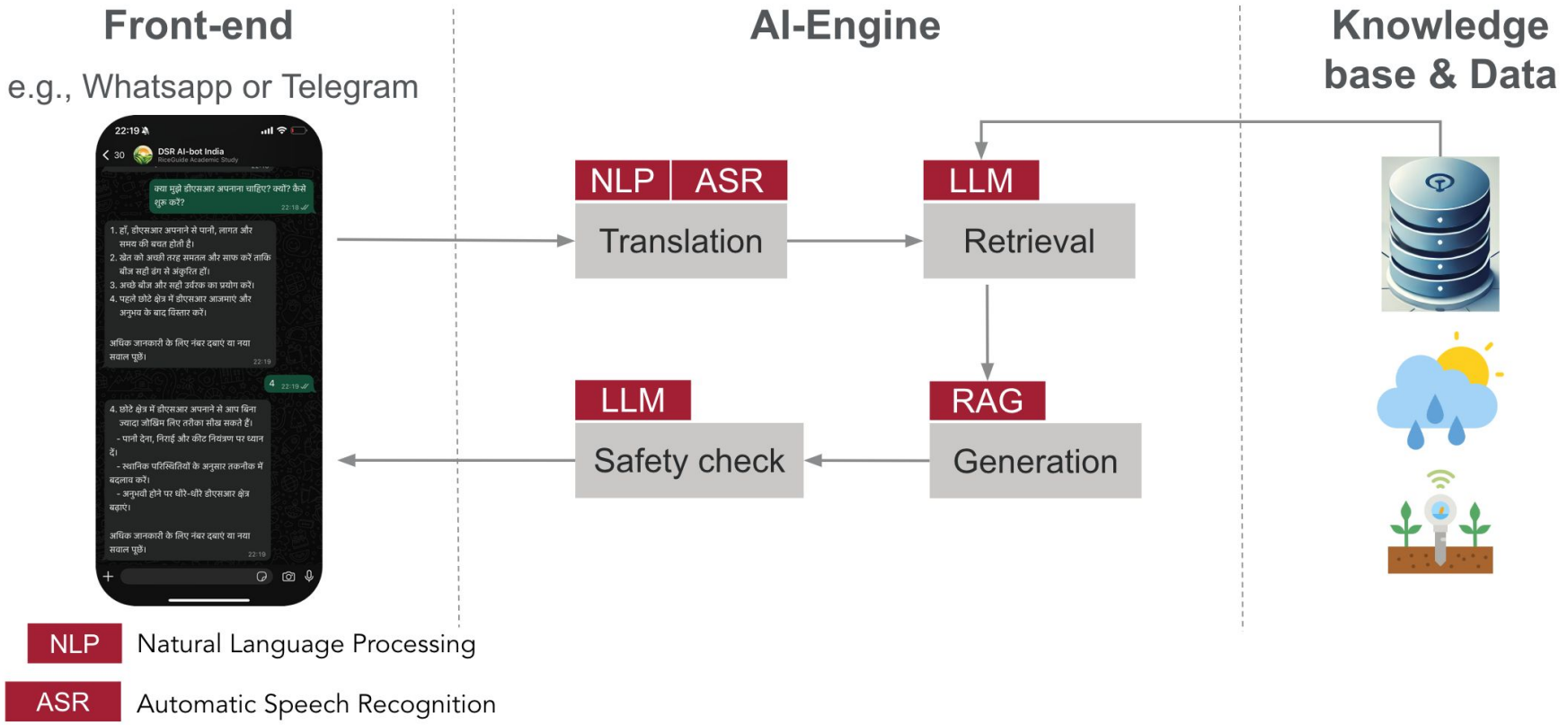
Upfront payment (T1) increases successful adoption among **highly educated farmers**

Table: Effect of Treatment 1 on Successful Adoption (vs control)

	Model 1 (No covariates)	Model 2 (Control for education)
Treatment 1	**0.175* (p=0.085)**	-0.030 (p=0.726)
Education	—	0.028 (p=0.118)
Education × Treatment 1	—	**0.051** (p=0.045)**

Observation: highly educated farmers see larger gains in success under Treatment 1.

This year: can AI assistant help close the gap?



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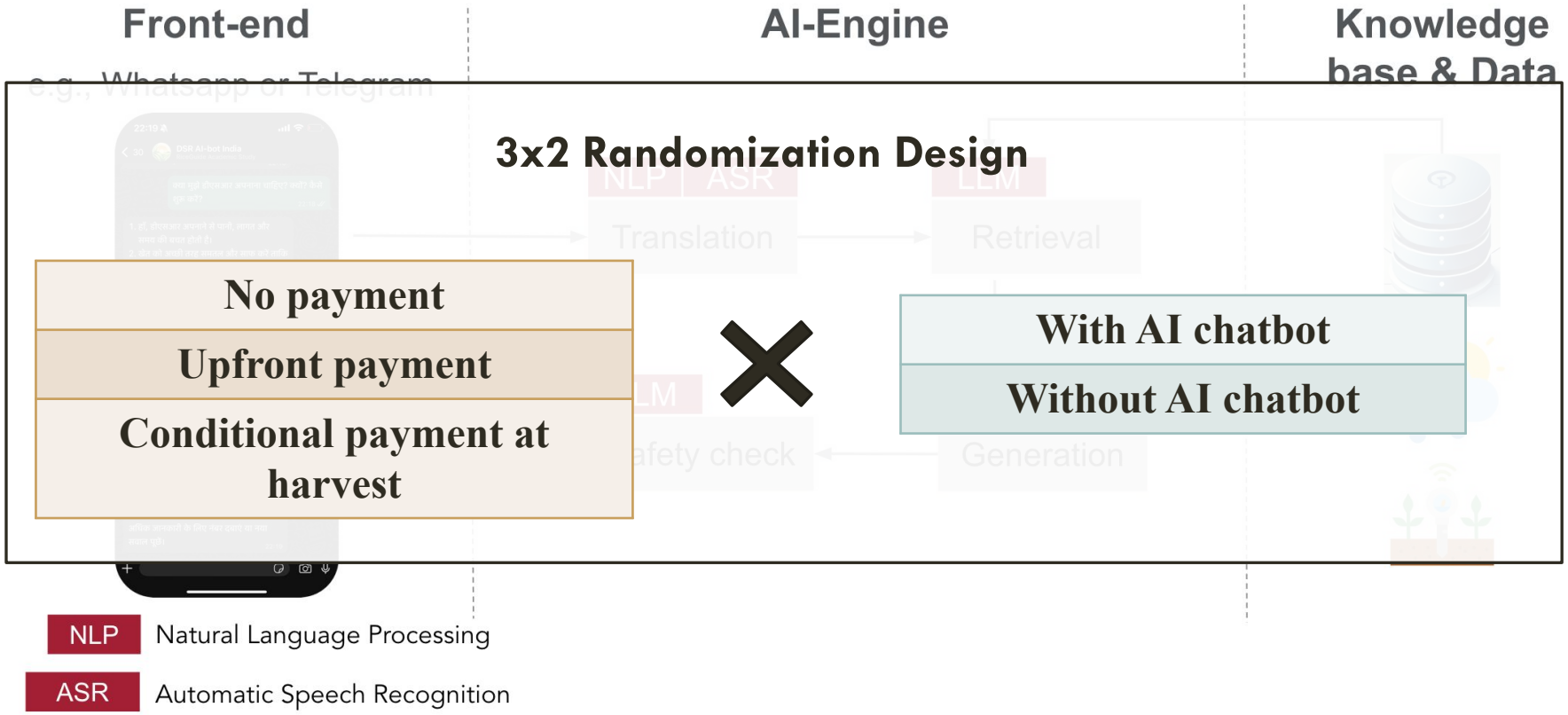
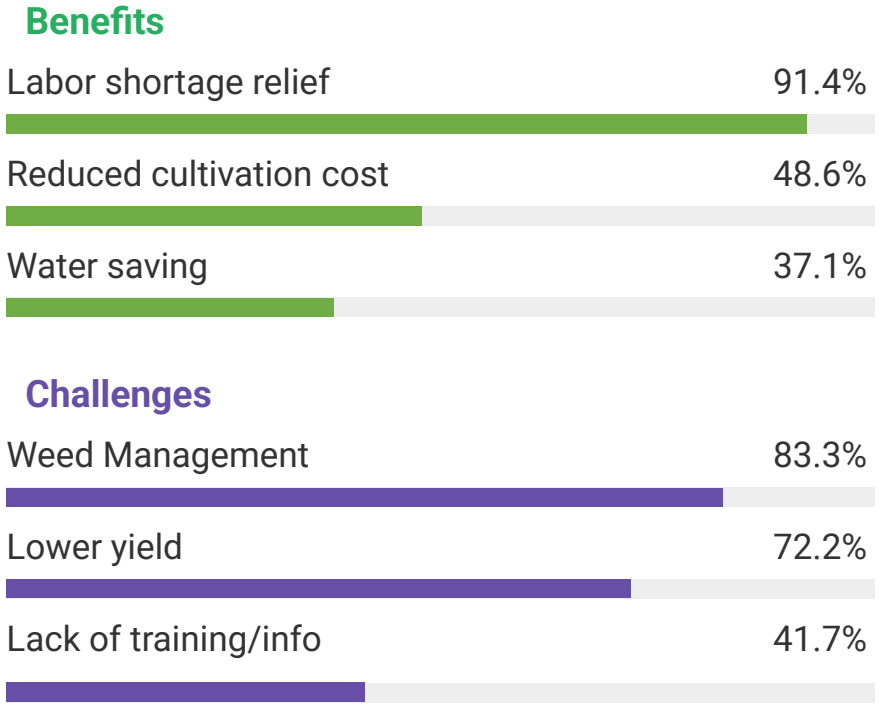


Challenges & Feedback

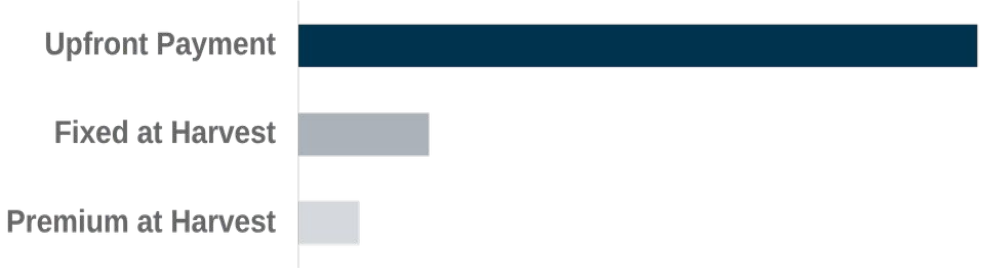
Why Non-Users Did Not Adopt?

What DSR Users Experienced?



Payment Incentive: Farmers favor **early upfront support,** but sustained DSR adoption may require higher payment levels or additional support.

Payment Preference Ranking

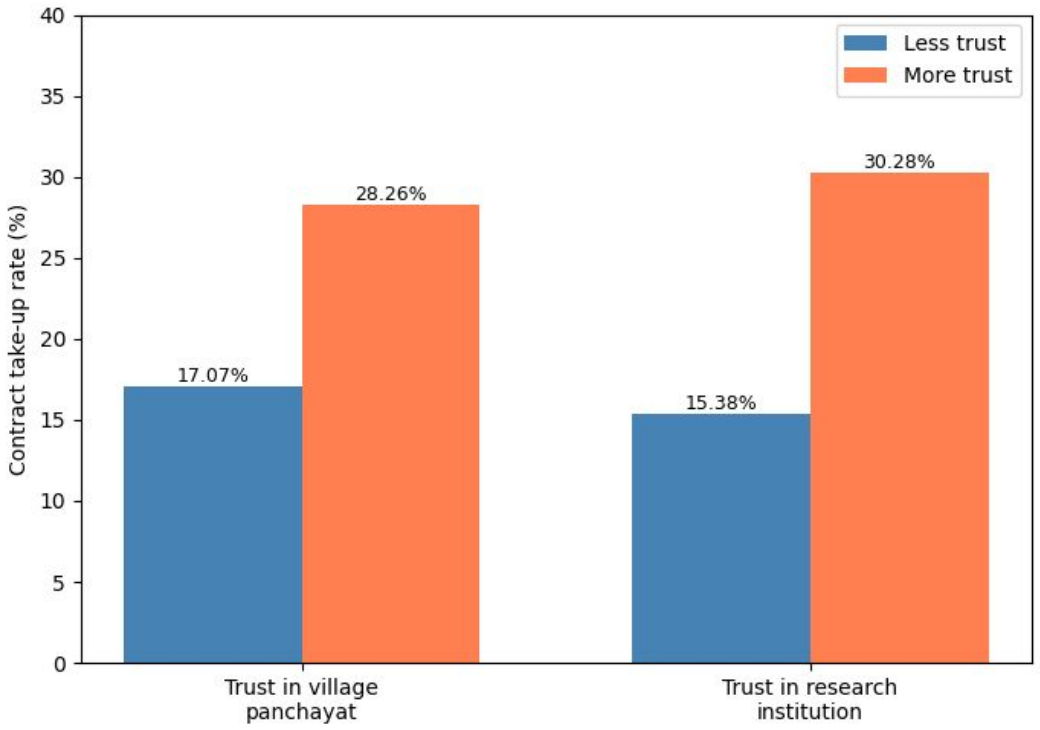


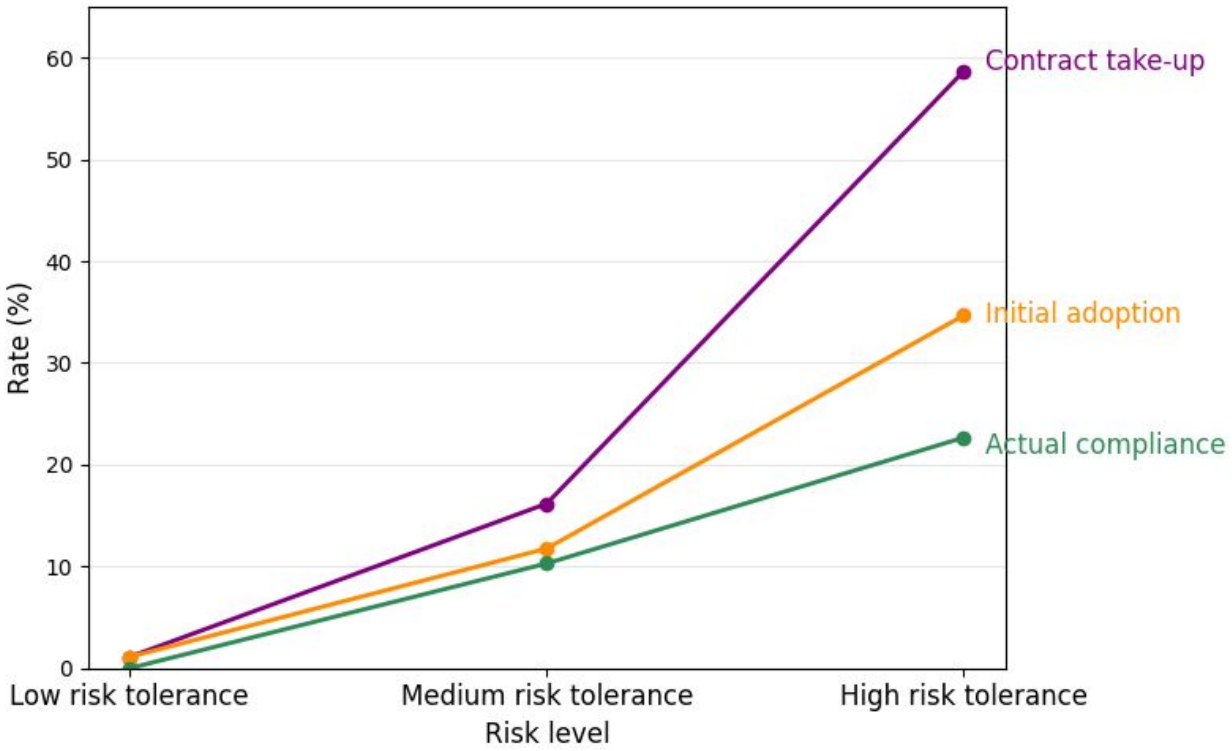
Timing Preference for Payments

Other Takeaways from the Current Pilot

- **Payments were primarily used for DSR-related inputs:** 68% for seeds and 24% for fertilizer.
- **Payment rules were generally clear:** 95% of respondents said the rules were easy to understand.
- **Indicated Support Needed** centers on financial support, technical training, and machinery access.

Beyond payment design, **institutional trust and farmers' willingness to bear risk** are linked to contract takeup and compliance





Farmers with higher trust in local governance and the research partner are more likely to sign the contract.

More risk-tolerant farmers are consistently more likely to participate and follow through.

Plan for this year: proposed scale

Control: 40 villages (20 with AI 20 without)
Upfront Payment: 40 villages (20 with AI 20 without)
Conditional Payment: 40 villages (20 with AI 20 without)

120 Villages in total, 15-20 farmers per village (~2000 farmers in total)

Plan for this year: proposed timeline

March: Finalizing experimental design and prepare recruitment materials

WRG/IRRI Input:

- Identify a list of districts with high density paddy cultivation
- Collect DSR guidance materials for AI bot development

April-May: Farmer recruitment in the field